



- (a) (i) Complete the trace table below by dry running the procedure `Place()` when it is called by the statement:

`CALL Place(1043, 2, 8)`

Value	Start	Unused	New	Last	Current

[4]

- (ii) Complete the diagram showing the contents of the global arrays `Data` and `Pointer` after the procedure `Place()` has run to completion when called as shown in part (a)(i).

	Data		Pointer
1	1018	1	
2	1007	2	
3	1010	3	
4	1056	4	
5	1092	5	
6	1062	6	
7	1034	7	
8		8	
9		9	
10		10	

[3]

- (b) The operation carried out by procedure `Place()` together with the arrays form part of the implementation of an Abstract Data Type (ADT).

Identify the ADT and state the operation carried out by procedure `Place()`

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..... [2]

